

# **AI/ML Capstone Project Completion Guide**

**Day 3 Task Update: Research Framing & Mentor Review**

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# Slide 1: Completed Work & Research Framing

## Refined Research Question:

*"For UWOC bit recovery, how does classification performance (accuracy, F1, and Bit Error Rate) of a fixed-threshold detector compare against Logistic Regression, SVM, and Random Forest classifiers, when evaluated separately across individual NTU turbidity levels rather than as one pooled average?"*

## Completed Day 1 & 2 Deliverables:

- **Day 1 Note Done:** Restated problem, established targets/success criteria, defined evaluation metrics, and outlined 9 core clarification questions.
- **Pipeline Diagram:** Built a clean, structured initial system pipeline mapping inputs to target outputs.
- **Day 2 Literature Review:** Completed comprehensive comparison table analyzing candidate baseline vs. machine learning methods.

## Slide 2: Measurable Outputs & Evaluation Plan

To rigorously evaluate the framework, our progress metrics are explicitly defined as:

- **BER Reduction (%):** Quantified error reduction achieved by the best-performing ML model versus the fixed-threshold baseline, evaluated distinctly for each individual NTU level.
- **Performance Matrix:** A structured comparison table tracking metrics (Accuracy, Precision, Recall, F1, BER, and Inference Time) across all proposed models and across all distinct NTU levels.
- **Degradation Thresholds:** Statistical identification of the specific turbidity limits (NTU levels) where the fixed-threshold baseline begins to fail significantly.
- **Optimized Recommendation:** One finalized model recommendation justified objectively by an accuracy-versus-complexity/latency tradeoff, rather than blindly choosing highest raw accuracy.

# Slide 3: Next Actions, Risks & Dependencies

## Current Blockers & Dependencies:

- **Dataset Receipt:** Real dataset arrival is delayed by a few days. Days 4-6 will actively utilize this waiting period to build robust, data-agnostic preparation artifacts.

## Immediate Next Plan (Days 4-6):

- **Day 4 (Dataset Design & Quality Expectation):** Define target schemas and establish a pre-emptive risk checklist focusing on data imbalance, missing variables, and data leakage.
- **Day 5 (Pipeline & Experiment Planning):** Set up repo directory structures, design clear experimental log templates, and formalize structural baselines.
- **Day 6 (Methodology V1 Consolidation):** Package all definitions, risks, schemas, and workflows into a comprehensive package ready for the next scheduled mentor defense.